

# Amine El Messaoudi

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## EDUCATION

### Carleton College | Northfield, MN

*Expected Jun 2028*

*B.A. in Computer Science, Mathematics | Minor: Data Science and Statistics*

**Relevant Courses:** Algorithms, Data Structures, Comp Systems, Software Design, Linear Algebra, Math Structures, Calculus 3

**Honors:** Carleton Underbrink Scholar — Ranked #1 in high school class of 400

### Aquincum Institute of Technology (BME), Study Abroad | Budapest, Hungary

*Aug 2026 – Dec 2026*

**Coursework:** Theory of Computing, Machine Learning, Data Science, Applied Cryptography, Applied AI, Entrepreneurship

## TECHNICAL SKILLS & LANGUAGES

**Programming Languages:** Python, C, C++, SQL, JavaScript, Java, Swift, R, HTML, CSS, x86-64 Assembly, PHP

**Tools & Frameworks:** Flask, PostgreSQL, MongoDB, SwiftUI, Git/GitHub, GDB, Unix/Bash, Pandas, OpenCV, YOLOv8,

TensorFlow/TensorFlow Lite, OpenAI API, Docker, Ollama, LibreChat

**Practices:** Agile, Test-Driven Development, MVC Architecture, REST API Design, Offline-First Architecture

**Languages:** French (native), Arabic (native), English (fluent), Spanish (elementary proficiency)

## PROJECTS

### Real-Time Drowsiness Detection System | Python, YOLOv8, OpenCV

*Dec 2025*

- Trained a custom YOLOv8 object-detection model on labeled fatigue data, tuning confidence thresholds to reliably detect eye closure in real-time demo conditions
- Built a real-time inference pipeline processing live webcam and video streams, triggering automated audio alerts within 1s of sustained drowsiness detection

### EcoPulse – Smart Energy Dashboard | Python, Flask, Pandas, OpenAI API

*Oct 2025*

- Engineered a full-stack dashboard that ingests and visualizes 90,000+ energy records, surfacing consumption anomalies and generating AI-driven sustainability reports per household
- Automated data cleaning and analysis pipelines with Pandas, handling missing values, type conversions, and deduplication across the records

### NBA Players' StatCompare | Python, Flask, PostgreSQL, JavaScript, D3.js

*Sep 2025*

- Shipped a full-stack analytics app in a 3-person team, enabling side-by-side comparison of 500+ NBA players across multiple statistical categories with interactive charts
- Designed parameterized SQL queries supporting dynamic filtering by player, season, and stat category across 6,000+ records with interactive D3.js visualizations

## PROFESSIONAL EXPERIENCE

### AI/Software Developer | Academic Technology Support, Carleton College | Northfield, MN

*Jun 2026 - Present*

- Deploy and maintain self-hosted LLM infrastructure (LibreChat, Ollama, and other open models) on local hardware, giving faculty and students privacy-preserving access to AI tools independent of third-party APIs
- Built Lizard Wizard, a three-tier iOS/Flask/MongoDB app for lizard re-identification using MobileNetV2 embeddings; worked across backend and iOS on a student developer team, with offline-first on-device TensorFlow Lite inference and deferred sync for fieldwork with no connectivity
- Built QuireMaker, a zero-dependency JavaScript tool that computes historical book-binding imposition layouts by simulating the physical folding process; deployed as a static site for Carleton's Book Studies and Special Collections courses

### Residential Assistant | Carleton College Residential Life Office | Northfield, MN

*Aug 2025 - Present*

- Led 10+ community initiatives (meetings, workshops, events) for a 50-resident community, using resident feedback surveys and iterative planning to increase participation and engagement
- Designed operational workflows for incident response, including escalation paths and resolution checklists, coordinating with multiple campus departments to resolve issues

### Office Assistant | Carleton College Registrar's Office | Northfield, MN

*Sep 2024 - Jun 2025*

- Managed and verified 8,000+ student and alumni records in the registrar database and processed transcript requests, correcting data-entry errors to maintain accuracy across registration cycles

### Data Analytics Intern | Global Salt Distribution Company | Fes, MOROCCO

*Jun 2022 - Sep 2022*

- Built pivot-table dashboards and analytical reports that surfaced sales trends and shipment inefficiencies, presenting findings directly to management
- Cleaned and validated 10,000+ sales, customer, and shipment records, ensuring data integrity for operational reporting

## LEADERSHIP EXPERIENCE

### President | INTERACT – Community Service Organization | Fes, MOROCCO

*Sep 2021 - Jun 2024*

- Led a 100+ volunteer organization and directed a relief campaign after the 2023 Morocco earthquake, mobilizing funds, clothing, and essential supplies for affected communities